

Chapter 04 – The Repetition Structure (Loops in JavaScript)

Full Exam Paper with Answers

Lecturer: Dr. Ahmed Taajir

Part I – Multiple Choice Questions (20 Marks)

1. 1. What is a repetition structure in programming?

- A) Executes code only once
- B) Executes code repeatedly
- C) Stops code execution
- D) Changes variable type

☒ Correct answer: B

2. 2. Which of the following is NOT a type of loop in JavaScript?

- A) for
- B) while
- C) repeat
- D) do...while

☒ Correct answer: C

3. 3. The number of times a loop executes is called:

- A) repetition
- B) iteration
- C) recursion
- D) repetition count

☒ Correct answer: B

4. 4. A loop that checks the condition before execution is called:

- A) Pre-test loop
- B) Post-test loop
- C) Infinite loop
- D) Sentinel loop

☒ Correct answer: A

5. 5. Which of the following is a pre-test loop?

- A) for
- B) while
- C) do...while
- D) Both A and B

☒ Correct answer: D

6. 6. Which loop guarantees at least one execution?

- A) for
- B) while
- C) do...while
- D) nested loop

☒ Correct answer: C

7. 7. The syntax for a for loop includes:

- A) initialization, condition, increment
- B) only condition
- C) only increment
- D) only initialization

☒ Correct answer: A

8. 8. What type of loop continues until a special value is entered?

- A) Infinite loop
- B) Sentinel loop
- C) Break loop
- D) Post-test loop

☒ Correct answer: B

9. 9. What keyword is used to exit a loop immediately?

- A) continue
- B) break
- C) stop
- D) exit

☒ Correct answer: B

10. 10. Which keyword skips the current iteration of a loop?

- A) end

- B) skip
- C) break
- D) continue

☒ Correct answer: D

11. 11. A loop that never stops executing is called:

- A) Pre-test
- B) Post-test
- C) Infinite loop
- D) Sentinel loop

☒ Correct answer: C

12. 12. The while loop tests the condition:

- A) Before execution
- B) After execution
- C) Never
- D) Twice

☒ Correct answer: A

13. 13. The do...while loop tests the condition:

- A) Before execution
- B) After execution
- C) At the start
- D) Randomly

☒ Correct answer: B

14. 14. In the for loop, the increment part:

- A) Decreases the variable
- B) Stops the loop
- C) Runs after each iteration
- D) Runs before each iteration

☒ Correct answer: C

15. 15. A loop within another loop is called:

- A) Infinite loop
- B) Nested loop
- C) Double loop

- D) Inner loop

☒ Correct answer: B

16. 16. Which of the following produces an infinite loop?

- A) while(true)
- B) for(i=0;i<5;i++)
- C) do{}while(false)
- D) None

☒ Correct answer: A

17. 17. A sentinel value is:

- A) An input to end a loop
- B) A loop counter
- C) A function
- D) A keyword

☒ Correct answer: A

18. 18. In a multiplication table using loops, which loop is inside?

- A) outer loop
- B) main loop
- C) inner loop
- D) top loop

☒ Correct answer: C

19. 19. The 'break' statement can be used in:

- A) for
- B) while
- C) do...while
- D) All of the above

☒ Correct answer: D

20. 20. The 'continue' statement is used to:

- A) Exit the loop
- B) Restart the loop
- C) Skip current iteration
- D) Stop program

☒ Correct answer: C

Part II – True or False (20 Marks)

1. A repetition structure executes a block of code repeatedly.

Answer: True

2. Loops reduce code duplication and automate tasks.

Answer: True

3. The for loop is a post-test loop.

Answer: False

4. The while loop checks condition before executing.

Answer: True

5. The do...while loop always runs at least once.

Answer: True

6. Iteration means the number of times a loop executes.

Answer: True

7. A pre-test loop checks the condition after execution.

Answer: False

8. A post-test loop checks the condition before execution.

Answer: False

9. A sentinel loop uses a special value to end the loop.

Answer: True

10. Infinite loops stop automatically after 10 iterations.

Answer: False

11. The break statement exits a loop immediately.

Answer: True

12. The continue statement ends the entire loop.

Answer: False

13. The continue statement skips to the next iteration.

Answer: True

14. Nested loops contain one loop inside another.

Answer: True

15. The for loop is used when number of iterations is known.

Answer: True

16. The while loop is condition-controlled.

Answer: True

17. The do...while loop executes even if the condition is false.

Answer: True

18. Nested loops are used for multi-dimensional data.

Answer: True

19. A sentinel value is part of valid data.

Answer: False

20. Loops can help generate multiplication tables.

Answer: True

Part III – Direct Questions (20 Marks)

1. Define a repetition structure.

Answer: A control structure that executes code repeatedly until a condition is met.

2. List three main types of loops in JavaScript.

Answer: For, While, and Do...While loops.

3. Differentiate between pre-test and post-test loops.

Answer: Pre-test checks condition before execution; post-test checks after.

4. Write syntax of a for loop.

Answer: `for(initialization; condition; increment){ code }`

5. Write syntax of a while loop.

Answer: `while(condition){ code }`

6. Write syntax of a do...while loop.

Answer: `do { code } while(condition);`

7. What is an iteration?

Answer: One complete execution of the loop body.

8. Explain what causes an infinite loop.

Answer: When the condition never becomes false or is missing.

9. What is a sentinel loop?

Answer: A loop that ends when a special sentinel value is entered.

10. Give an example of a sentinel value.

Answer: -1 used to stop user input loop.

11. Define break statement.

Answer: A statement used to exit a loop immediately.

12. Define continue statement.

Answer: A statement that skips the rest of current iteration.

13. What is a nested loop?

Answer: A loop inside another loop.

14. Give one example of nested loop usage.

Answer: Generating multiplication tables.

15. Write example code for a multiplication table using nested loops.

Answer: `for(i=1;i<=10;i++){ for(j=1;j<=10;j++){ console.log(i*j); } }`

16. Differentiate between while and do...while loops.

Answer: while checks before; do...while checks after running code once.

17. Explain count-controlled vs condition-controlled loops.

Answer: Count-controlled runs fixed times; condition-controlled depends on a condition.

18. What is the purpose of loop control statements?

Answer: To manage the execution flow using break or continue.

19. What happens if a loop lacks increment or condition?

Answer: It becomes an infinite loop.

20. State two benefits of using loops.

Answer: Automation of tasks and code reusability.